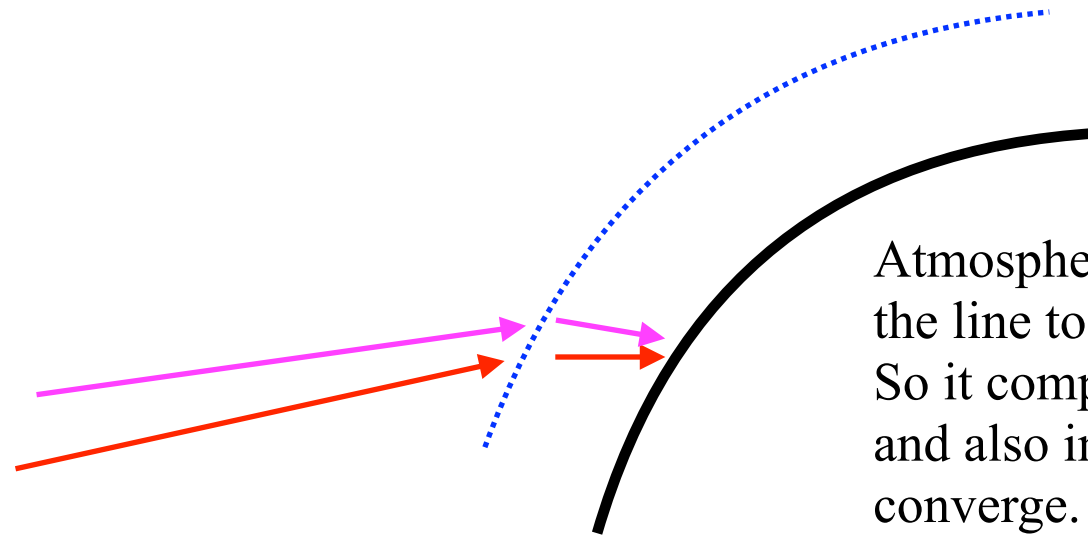


Differential atmospheric refraction and image motion for DESI

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DESI-9817 v2



Atmospheric refraction pushes objects up along the line to zenith, and is greater at low elevation. So it compresses the field in the “up” direction, and also inward because the lines to zenith converge. The compression is $\sim$ few arcsec over a DESI field radius 1.6 deg. Depends on altitude and field rotation.

Resources:

An ADC does not compensate for this effect!

M. Lampton, DESI-4957 - Sky distortions and rotations

D. Kirkby, DESI-8586 - High airmass guiding

J. Meeus, “Astronomical Algorithms,” chap. 16

R.C. Stone, 1996, PASP, 108, 1051

J-G. Cuby et al, 1998, SPIE, https://www.eso.org/sci/facilities/paranal/decommissioned/vimos/inst/atmos/atmos_paper.pdf

R. Sanchez-Janssen et al, 2014, A&A, 566, 2

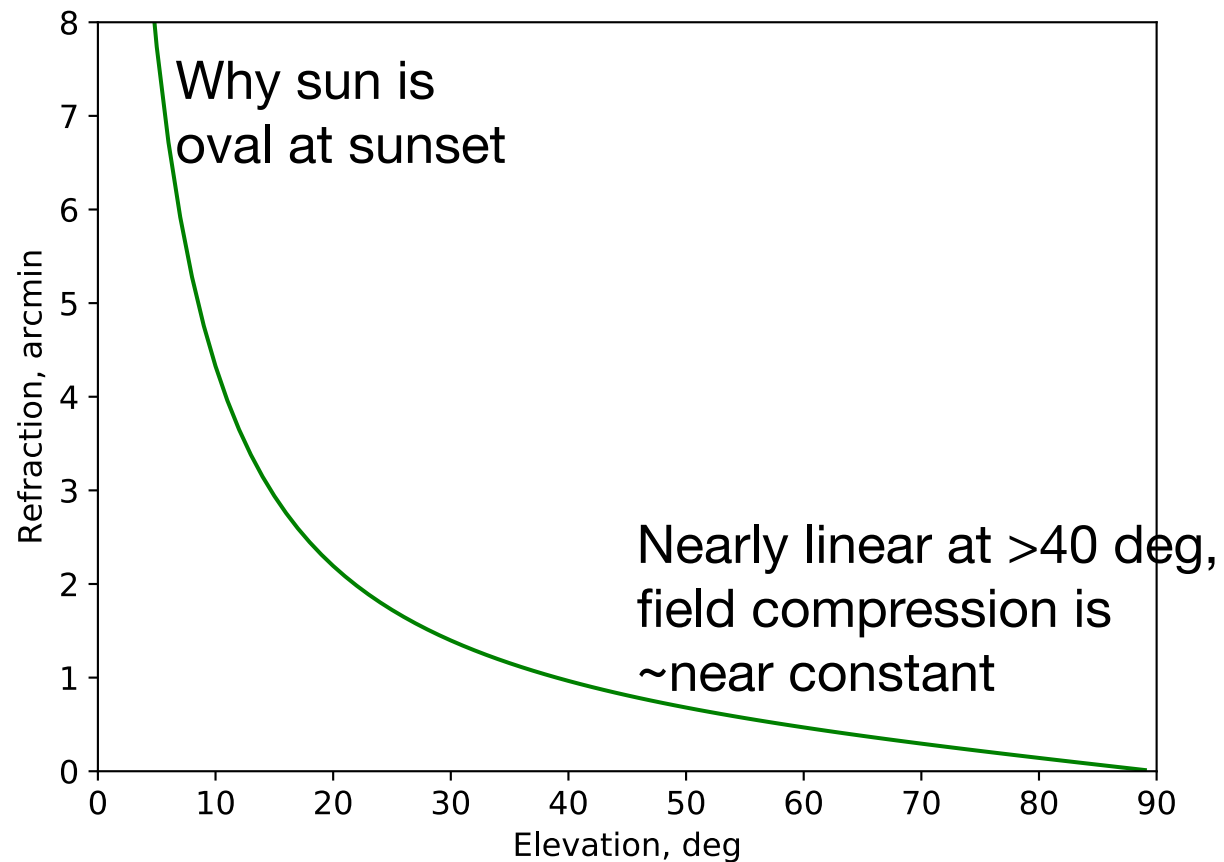
https://github.com/bjweiner/MMT/blob/master/binospec/atmo_refraction_desi.py

https://github.com/bjweiner/MMT/blob/master/binospec/area_refraction.py

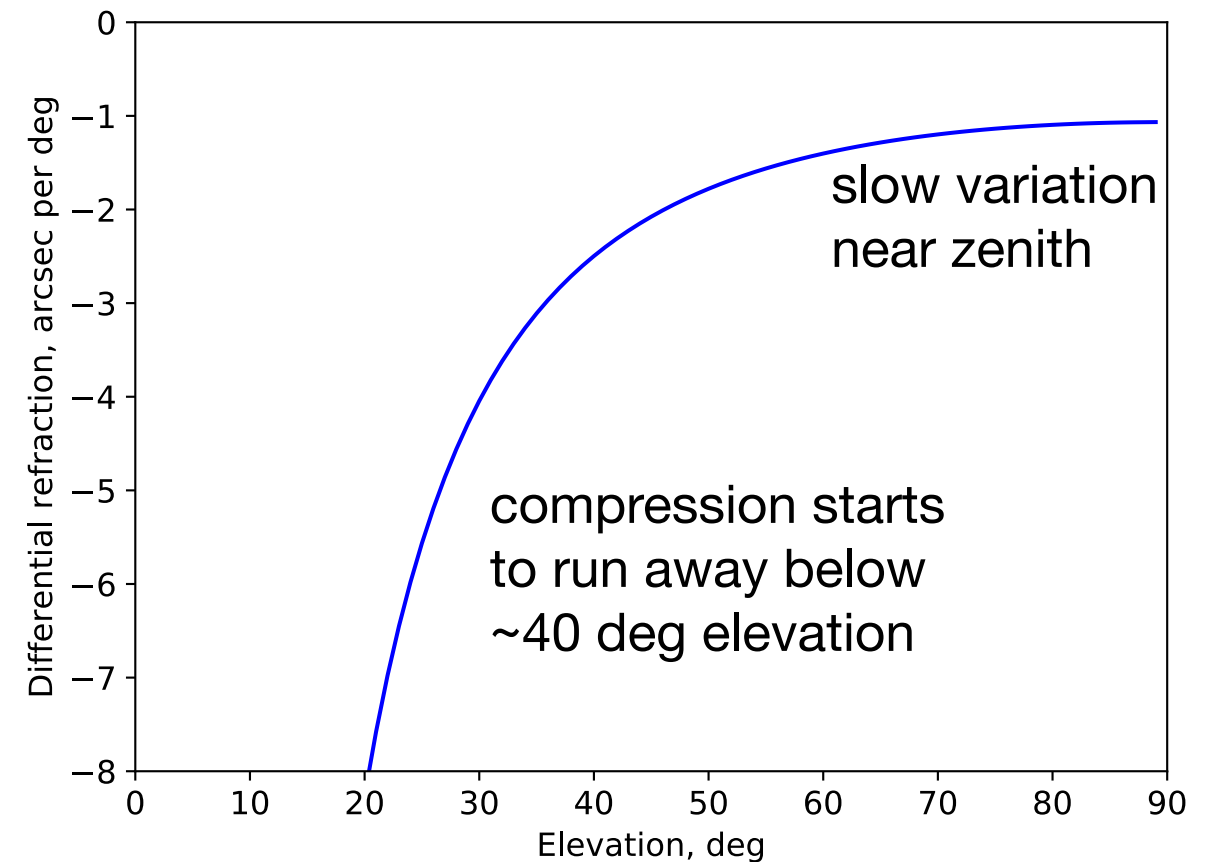
S. BenZvi, Dither test summary, https://docs.google.com/presentation/d/1MBGKtmInKCtjvBQgM84NIAoCa4rk_PohOCFsQsMUi3c/edit?slide=id.g36beef17f25_0_0

“Many problems in astronomy are caused by the fact that an arcsecond is really small.”

Total amount of the refraction isn't critical,
as long as you know where to point.
But differential across the field matters.



Derivative of refraction, in arcsec/deg,
gives the compression across a field



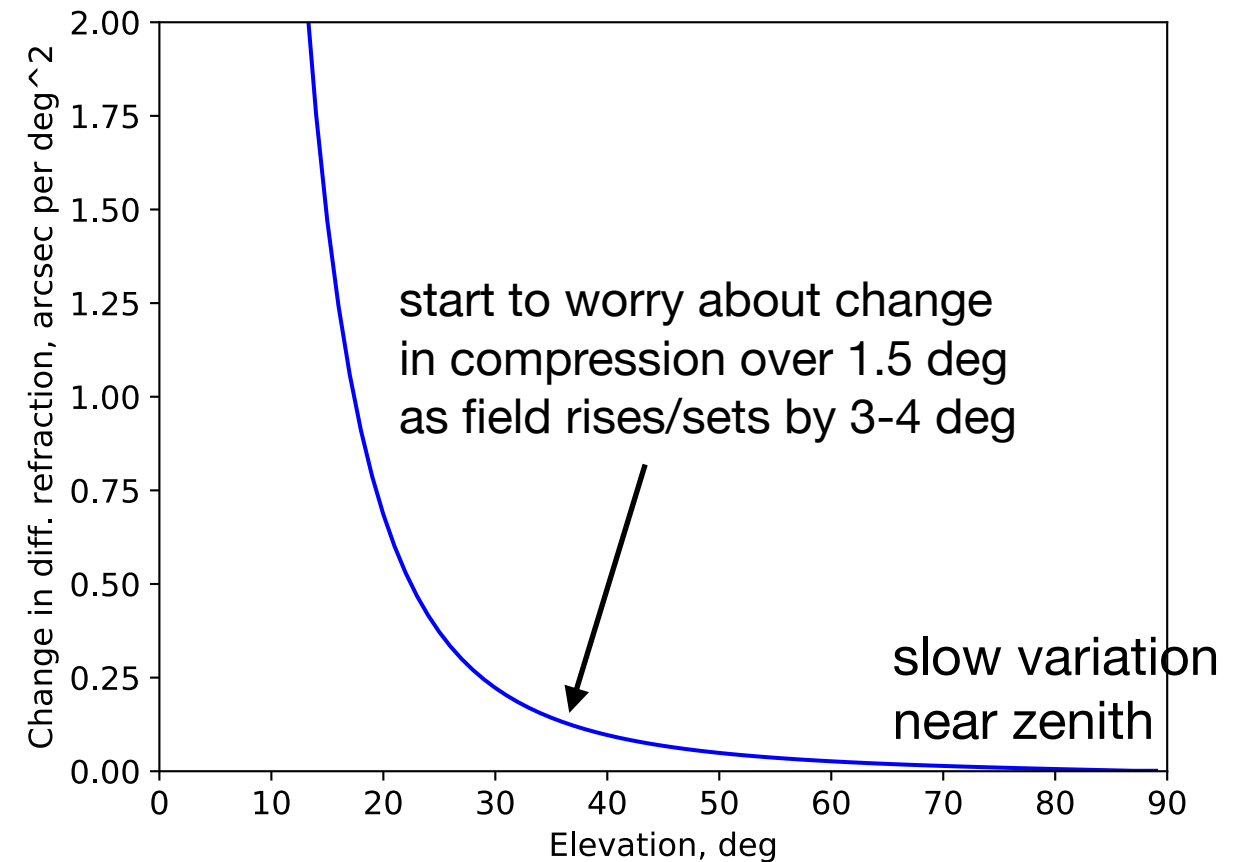
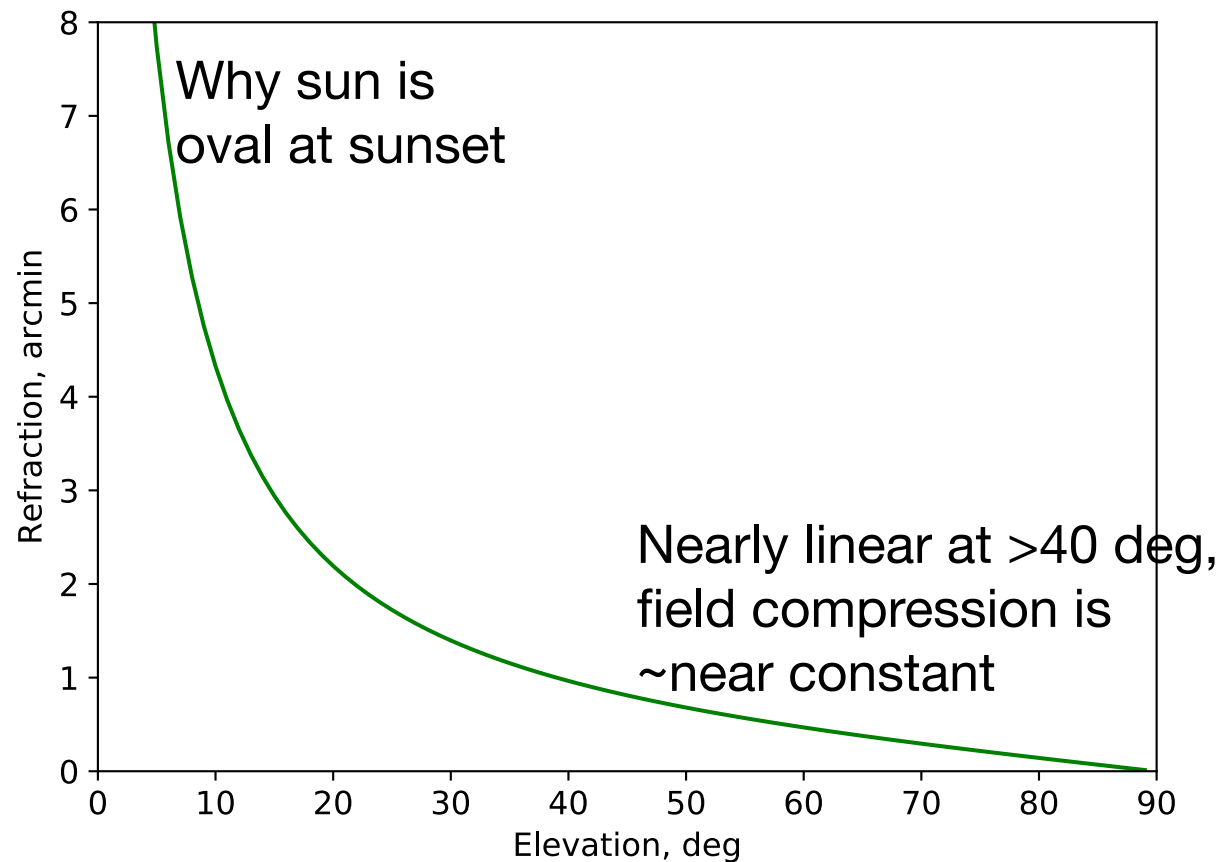
Typical compression fraction (Lampton DESI-4957) at 0.7 microns:

elev 30	d_alt 880 ppm,	over 1.5 deg radius: 4.8"	, and d_az 220 ppm
elev 60	293 ppm	1.6"	220 ppm
elev 90	220 ppm	1.2"	220 ppm

compression: $d_{\text{alt}} \sim 220 \text{ ppm} * \csc^2(\text{alt}) \sim 220 \text{ ppm} \sec^2(\text{zenith angle})$
 $2e-4 (\text{airmass})^2$ (Sanchez-Janssen et al 2014)

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Second derivative of refraction, in arcsec/deg²,
how the compression changes as a field moves



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What is the problem of differential spatial refraction?

Standard packages, software, telescopes convert true position to apparent position, so that you know where to point.

For multi-object spectroscopy, where to put fibers or slits depends a little on the differential refraction, both on the field altitude and orientation wrt zenith. If you don't have a very large field and/or you observe at high elevation, this is a fairly small effect (eg ~ 0.2 - $0.4''$ across 15 arcmin), so a slitmask design or fiber placement may design for a "typical" observation and not worry about changes in field orientation.

For DESI, with field radius of 1.6 deg and small fibers, differential apparent position clearly matters, and fiber placement is computed at time of observation. This works at high elevation, but tests at low elevation have suggested rapid degradation of S/N during the exposure, due to apparent motion. [*]

Tests of Low-Dec observing with dither sequences have suggested a coherent pattern of objects being offset from their expected positions (see slides from Segev BenZvi), and it's not clear if this is pointing to a problem in PlateMaker or something else.

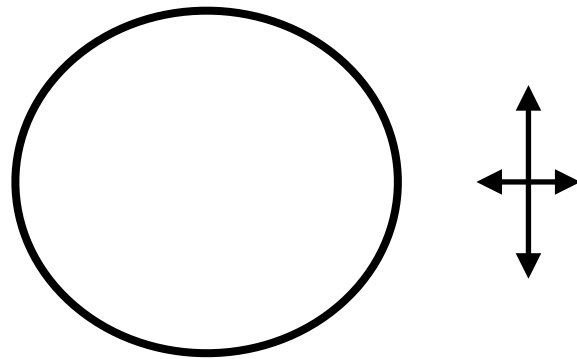
Low-Dec observing will be more pressing for DESI-Ib, so it will be useful to predict the size and patterns of apparent motion *during* an exposure.

People know this effect exists, but not many detailed writeups in the literature on it, vs. atmospheric dispersion. See Sanchez-Janssen et al (2014, A&A, 566, 2) for some approximate formulae on diff. refraction.

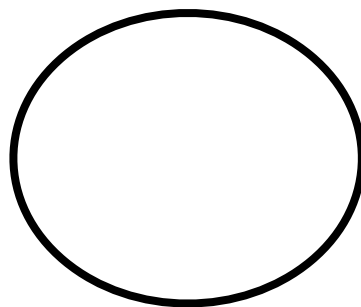
[*] I happened to be support observer during a low-elevation test in Sept 2024, and there was an extensive Slack discussion in #observing about it, but am not familiar with the entire history of these tests.

Effects of field motion:

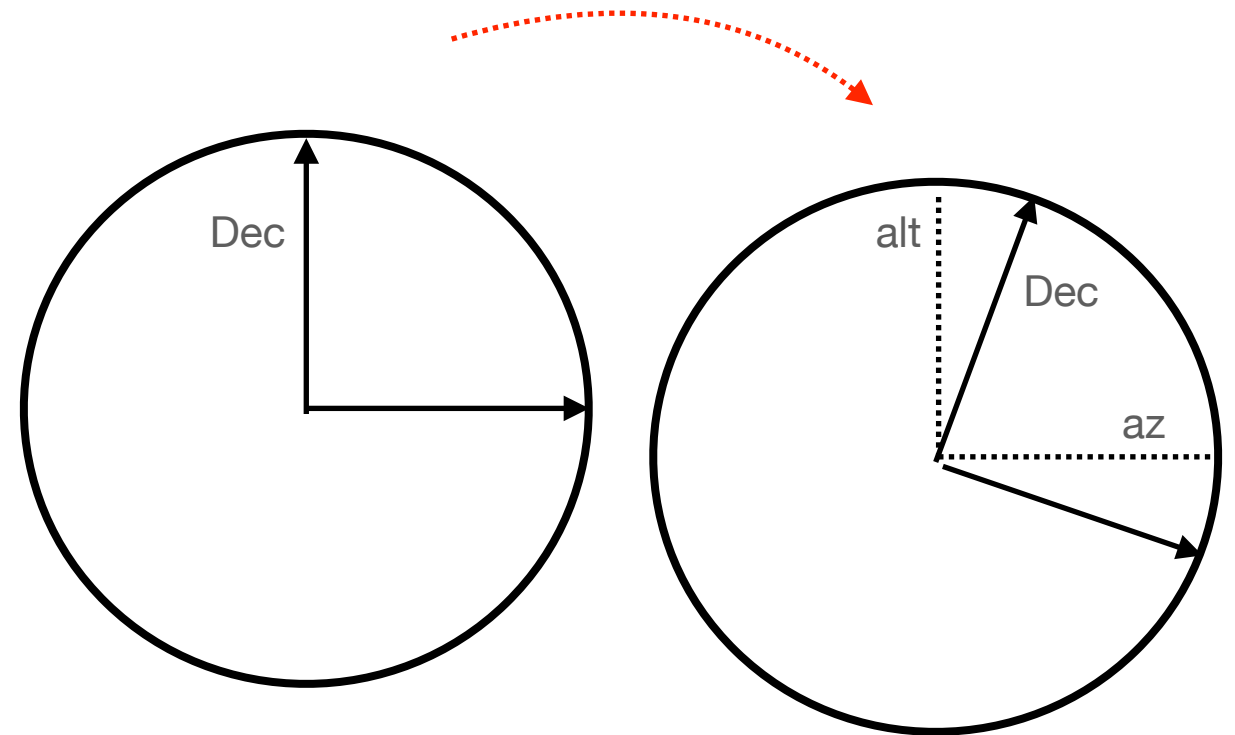
1. Rising / setting.
2. Field rotation changes altitude slightly, also changes direction of compression.



Rising: expands in up direction and slightly in across



Significant if field is low, but easy to mitigate by observing at alt > 45 or so.



Field rotation: the az-alt components change for a vector from center to a point at edge of field. Because az and alt are compressed by different amounts, the vector to the edge point changes. The point near 3 o'clock moves off in altitude by $\text{delta-alt} \sim R * \sin \theta * (d_{\text{alt}} - d_{\text{az}})$

e.g. for 4 deg field rotation at elev 30 deg,
 $\text{delta-alt} \sim 5400'' \times 0.07 \times (880 - 220)e^{-6}$
 $\text{delta_alt} \sim 0.25''$ at edge of DESI field.

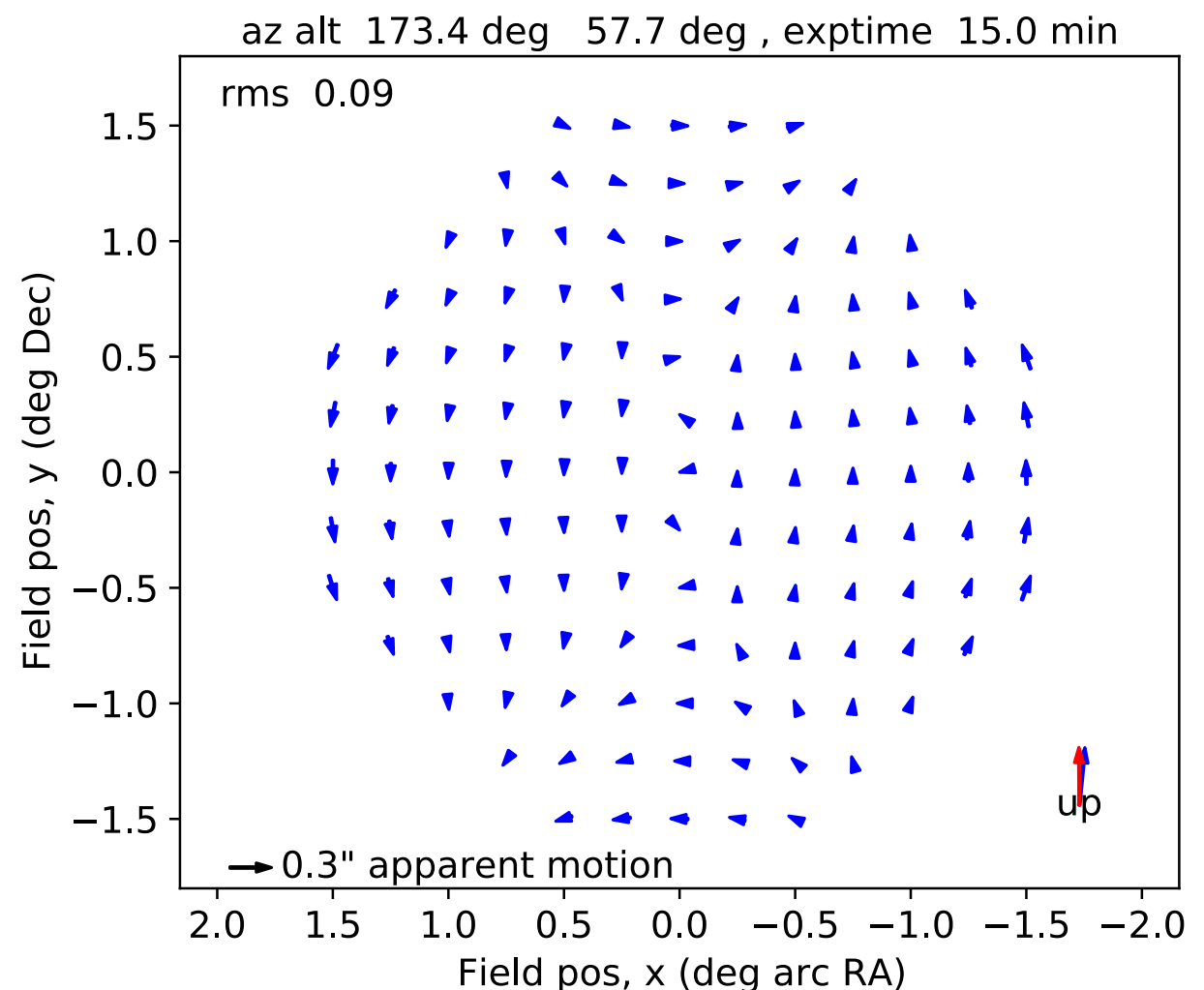
Computing refraction induced distortions during an exposure

use `astropy.coordinates`:

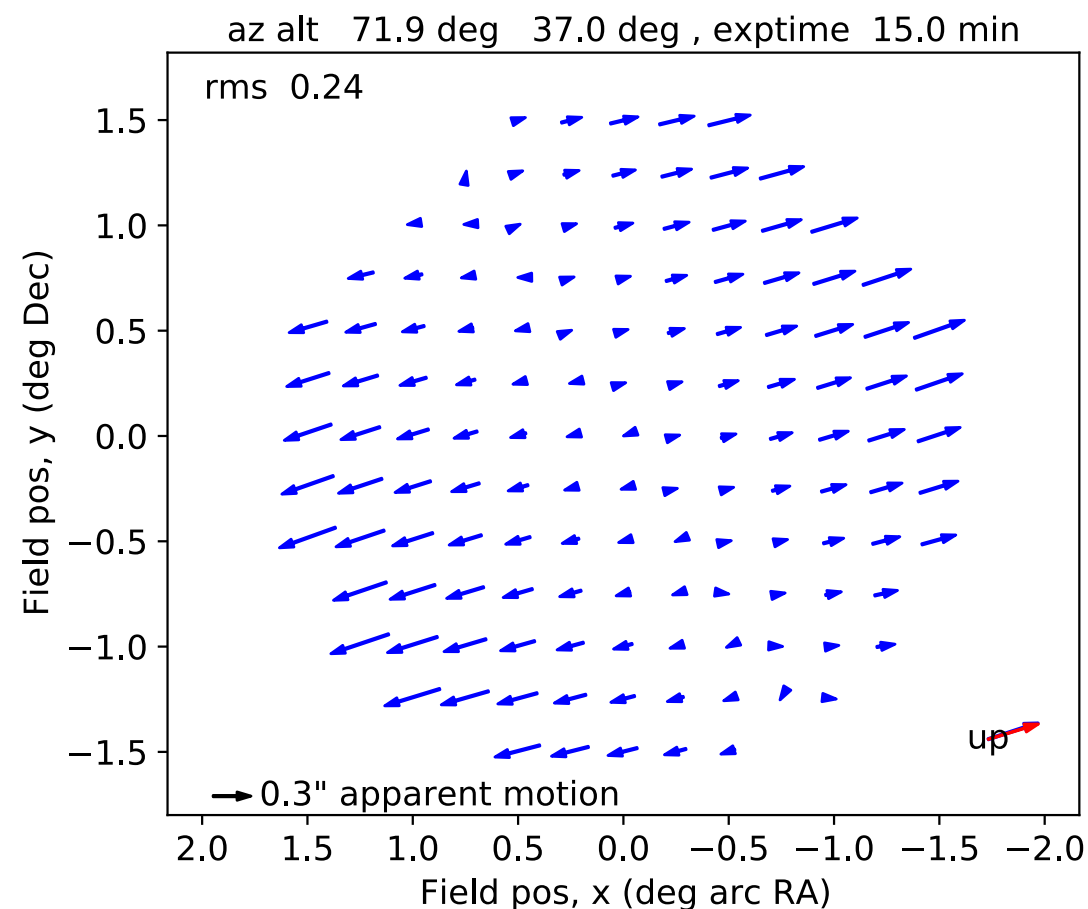
1. create a grid of points in true RA, Dec around a field center - `SkyCoord()`
2. convert to apparent az, alt grid at a given time (pressure, wavelength) - `.transform_to()`
3. get spherical offsets (u_1 , v_1) in arc units in az, alt, from apparent center to apparent grid points - `.spherical_offsets_to()`
4. derotate (u_1 , v_1) to (x_1 , y_1) apparent offsets in arc units from field center, ~~aligning y with the Dec axis~~, by the parallactic angle - own code
5. apparent grid at a second time (eg 15 min later) yields u_2 , v_2 offsets from a different az, alt apparent center
6. derotate (u_2 , v_2) to (x_2 , y_2) apparent offsets in arc units, ~~aligned to Dec~~, by the parallactic angle - own code
7. compute shift from x_1, y_1 to x_2, y_2 - these are both referenced to the field center and to Dec. - own code

We assume that the telescope tracks apparent field center. For version 2, we derotate by the parallactic angle (where v_1 referenced both fields to Dec axis). (The difference between v_2 and earlier v_1 plots is that aligning to the Dec axis introduced a small rotation.)

Field on the equator near transit, shows negligible shifts in 15 min exposure.
(Getting near-zero for an answer is a code test, since the apparent field has moved and rotated.)



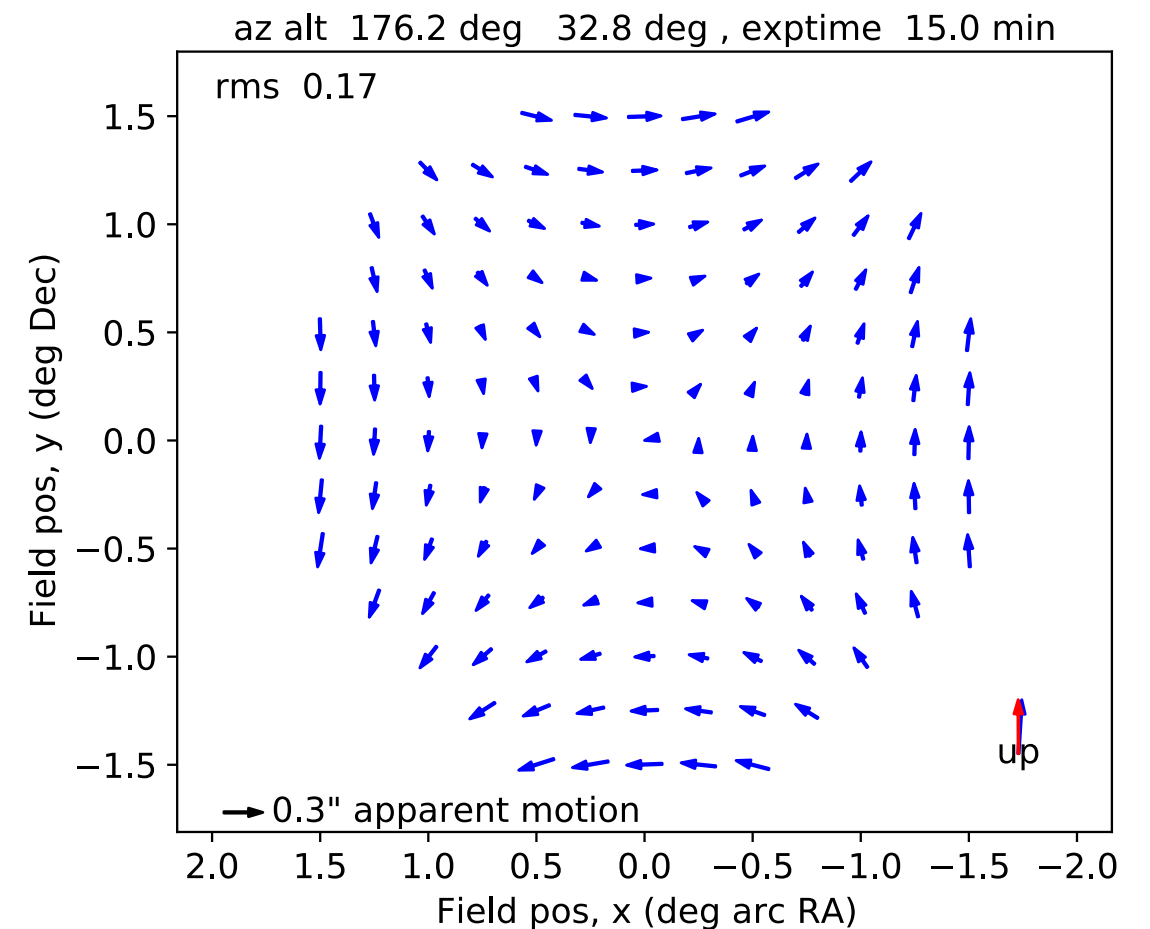
Rising field



Field that is rising, 4 hours east and dec +32, expands a lot in 15 minutes. But this is not hard to avoid by observing at higher alt. Length of arrows is greatly exaggerated, see the scale bar.

In the v2 version of this plot, derotating by the parallactic angle, the expansion is more closely aligned with the zenith direction. In v1, aligning zero rotation along the y (Dec) axis introduced a little spurious field rotation (since there is a little real distortion along the y-axis, shown in this plot).

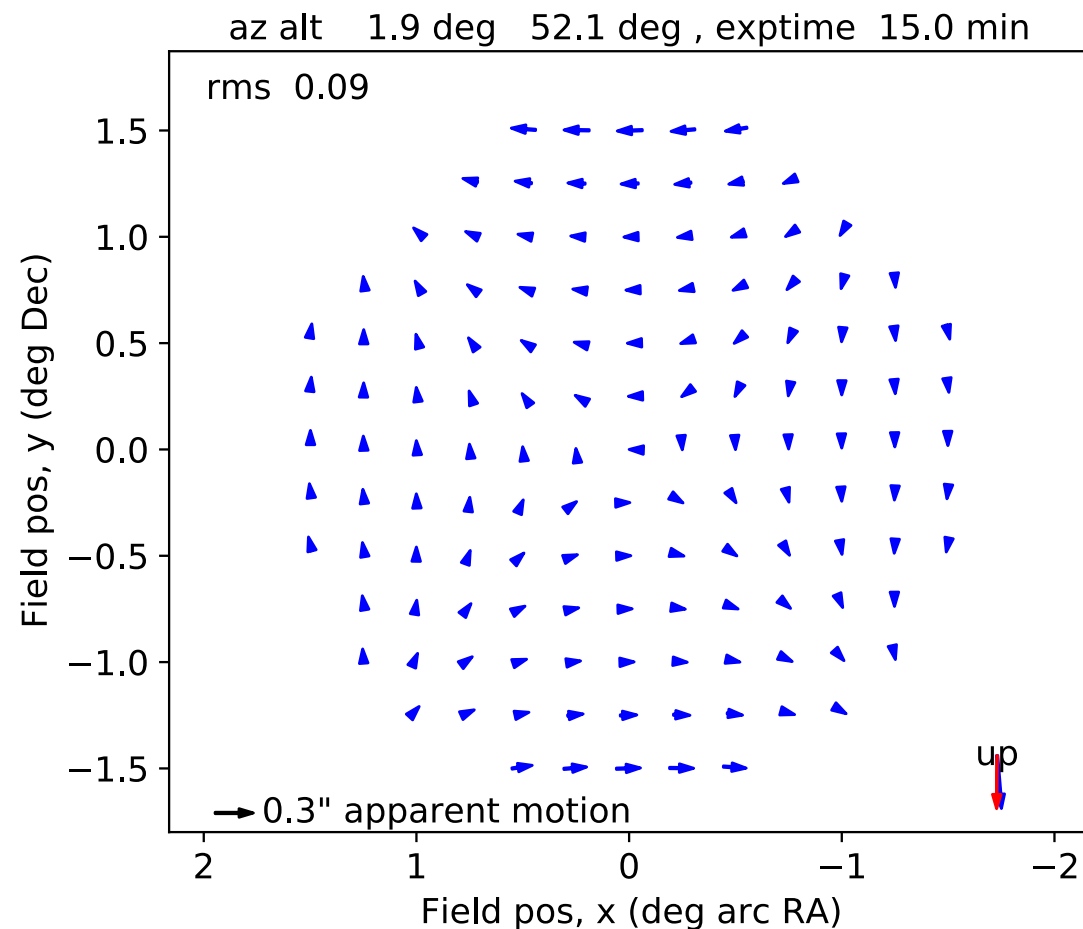
Low-Dec field near transit



Field at Dec -25 near transiting, shows significant image motion at each edge in 15 minutes, not easily avoidable. In v2, using the parallactic angle, the distortion is quadrupole (because the y-axis is not pinned to zero motion), as some commenters on v1 expected.

Does not look like the more-horizontal pattern observed in the Low-DEC dither tests. But it does have shear. How this manifests during a dither sequence may depend on how the guiders react to the image motion.

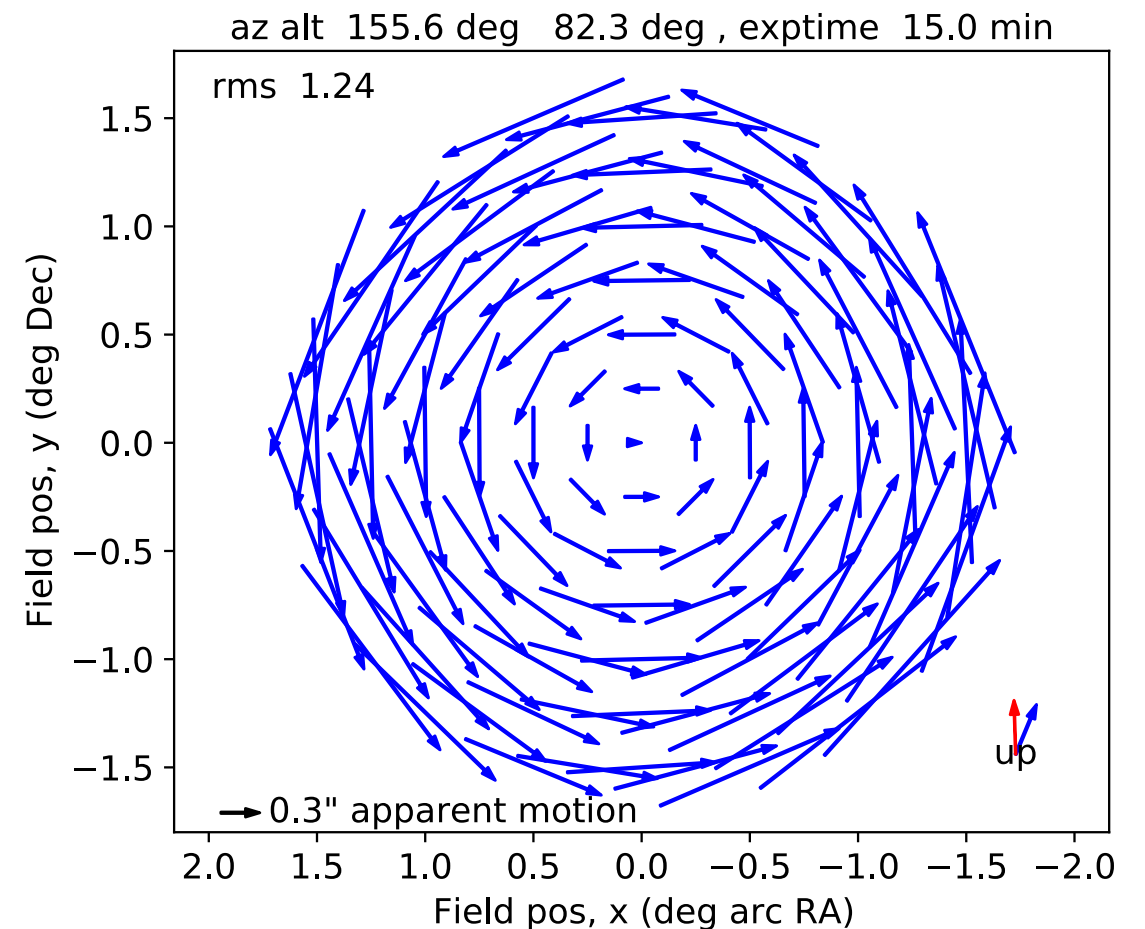
Northern field



Near transit at Dec +70, with moderate elevation, there is little apparent image motion.

In v2, using the parallactic angle to derotate, the amount of image motion is similar but now it has the expected quadrupole pattern.

Close to zenith



Distressingly, derotating with the parallactic angle improved the results in v2 *except* near the zenith, where it predicts substantial field rotation that can't be real, larger than in the v1 plot. (Near the zenith, the az and alt compression should be fairly similar, so field rotation shouldn't cause much quadrupole).

I am not sure of the reason, unless it is limited accuracy in the parallactic angle calculation (which is using a standard formula from Meeus that everyone uses).

Lessons / future study

Size of effects:

Differential refraction can have a large effect under certain conditions, mostly low elevation < 40 deg.

Field rotation has an effect, so estimating the shifts solely on elevation is not entirely reliable.

Observing at the meridian doesn't fully solve the problem; at low dec there is still a coherent shift pattern as the field transits. May need to mitigate this by limiting all the low-Dec exposures to $\sim 10 - 15$ minutes.

The spatial coherence of the pattern is a problem - if the positioning spec is $0.4''$, randomly being off by $0.4''$ is not as bad as having each edge of each field be off by $0.4''$.

Shape of effects:

The distortion field caused by differential refraction is not very intuitive. I was expecting mostly stretching along the altitude axis and more or less 0 at the midplane, and that's not always right, mostly due to field rotation. In the v2 plots, we can see that the distortion near transit is largely a quadrupole (see also D. Kirkby DESI-8586).

The patterns don't immediately explain the low-Dec dither sequence results, but need to think about how a coherent shift interacts with the guiders. These may be a framework for simulating the effect of different types of error in fiber positioning (random, guider-induced, simulate effect of different types of software bugs).

Caveats:

In v1, the de-rotation calculation by rotating the start and end positions back to relative-to-the-Dec-axis was a shortcut, and not really how an equatorial telescope tracks. v2 fixed this by derotating with the parallactic angle. That made everything better except the near-zenith plot (but we know that observing near the zenith works).

I don't trust the results near zenith, which probably has to do with the large field rotation, or possibly something about the offsets along az-alt axes produced by astropy near the zenith. Further thoughts on this are welcome.

I had to work through several small-angle effects that can screw things up, before I got to the point where observations at moderate elevations have only small shifts, as expected.